$\therefore \quad T_n = a + (n-1)d = 1 + 8(n-1) = (8n-7)$.

$8n - 7 = 1994 \Rightarrow 8n = 2001 \Rightarrow n = 250$.

$T_{250} = 1 + (250-1) \times 8 = 1 + 249 \times 8 = 1993$.

So, 1993 lies on thumb and 1994 on index finger.

171. Out of four consecutive integers two are even and therefore, their product is even and on adding 1 to it, we get an odd integer. So, n is odd. Some possible values of n are as under:

$n = 1 + (1 \times 2 \times 3 \times 4) = (1 + 24) = 25 = 5^2$,

$n = 1 + (2 \times 3 \times 4 \times 5) = (1 + 120) = 121 = (11)^2$,

$n = 1 + (3 \times 4 \times 5 \times 6) = (1 + 360) = 361 = (19)^2$,

$n = 1 + (4 \times 5 \times 6 \times 7) = 841 = (29)^2$ and so on.

Hence, n is odd and a perfect square.

172. $(x^2 + y^2) = (x+y)^2 - 2xy = (15)^2 - 2 \times 56$

$= (225 - 112) = 113$.

173. $(2^2 + 4^2 + 6^2 + \ldots\ldots + 40^2) = (1 \times 2)^2 + (2 \times 2)^2 + (2 \times 3)^2 + \ldots + (2 \times 20)^2 = 2^2 \times (1^2 + 2^2 + 3^2 + \ldots\ldots + 20^2)$

$= (4 \times 2870) = 11480$.

174. $5^2 + 6^2 + \ldots\ldots + 10^2 + 20^2 = (1^2 + 2^2 + 3^2 + \ldots\ldots + 10^2) - (1^2 + 2^2 + 3^2 + 4^2) + 400$

$= \frac{1}{6} n(n+1)(2n+1) - (1 + 4 + 9 + 16) + 400$, where $n = 10$

$= \left(\frac{1}{6} \times 10 \times 11 \times 21\right) - 30 + 400 = (385 - 30 + 400) = 755$.

175. $(6 + 12 + 18 + 24 + \ldots + 60) = 6 \times (1 + 2 + 3 + 4 + \ldots\ldots + 10) = 6 \times 55 = 330$.

176. $2^m > 960$ when the least value of m is 10.

Then, $2^{10} = 1024$ and $1024 - 960 = 64 = 2^6$.

$\therefore$ $m = 10$ and $n = 6$.

177. We keep on dividing 33333... by 7 till we get 0 as remainder.

```
   47619
7)333333(
  28
   53
   49
    43
    42
     13
      7
     63
     63
      ×
```

$\therefore$ Required number = 47619

178. We keep on dividing 99999.... by 7 till we get 0 as remainder.

```
  142857
7)999999(
  7
  29
  28
   19
   14
    59
    56
     39
     35
      49
      49
       ×
```

$\therefore$ Required number = 142857.

Number of digits = 6.

179.

```
       56
987) 559981(
     4935
      6648
      5922
       7261
       2961
         ×
```

Clearly, 7261 must be replaced by 2961, which is possibvle if 6648 is replaced by 6218, which in turn is possible if 5599 is replaced by 5556.

Thus, the correct number is 555681.

180. Clearly, multiples of 2 and 5 together yield 0.

Since the product of odd numbers contains no power of 2, so the given product does not give 0 at the unit place.

181. Let $N = 1 \times 2 \times 3 \times 4 \times \ldots\ldots \times 1000 = 1000!$

Clearly, the highest power of 2 in N is very high as compared to that of 5.

So, the number of zeros in N will be equal to the highest power of 5 in N.

$\therefore$ Required number of zeros

$= \left[\frac{1000}{5}\right] + \left[\frac{1000}{5^2}\right] + \left[\frac{1000}{5^3}\right] + \left[\frac{1000}{5^4}\right]$

$= 200 + 40 + 8 + 1 = 249$.

182. Let $N = 10 \times 20 \times 30 \times \ldots\ldots \times 1000 = 10^{100} \times (1 \times 2 \times 3 \times 4 \times \ldots\ldots 100) = 10^{100} \times 100!$

Number of zeros in 100 ! = Highest power of 5 in 100 !

$= \left[\frac{100}{5}\right] + \left[\frac{100}{5^2}\right] = 20 + 4 = 24$.

$\therefore$ Number of zeros in $N = 100 + 24 = 124$.

183. Let $N = 5 \times 10 \times 15 \times 20 \times 25 \times 30 \times 35 \times 40 \times 45 \times 50$

$= 5^{10} \times (1 \times 2 \times 3 \times 4 \times \ldots\ldots \times 10) = 5^{10} \times 10!$

Highest power of 2 in 10! $= \left[\frac{10}{2}\right] + \left[\frac{10}{2^2}\right] + \left[\frac{10}{2^3}\right] = 5 + 2 + 1 = 8$.

Highest power of 5 in 10! $= \left[\frac{10}{5}\right] = 2$.

$\therefore$ $N = 2^8 \times 5^{12} \times k$.

Since highest power of 2 is less than that of 5, so required number of zeros = 8.

184. Clearly, highest power of 2 is much higher as compared to that of 5 in 60!, so

Required number of zeros = Highest power of

$5 = \left[\frac{60}{5}\right] + \left[\frac{60}{5^2}\right] = 12 + 2 = 14$.

185. Let $N = (1 \times 3 \times 5 \times 7 \times \ldots \times 99) \times 128$.

Clearly, N contains 10 multiples of 5 (5, 15, 25, 35, ..., 95) and only one multiple of 2 i.e. 128 or 2^7.

Clearly, highest power of 5 in N is greater than that of 2.

$\therefore$ Number of zeros in N = Highest power of 2 in $N = 7$.

186. $N = 2 \times 4 \times 6 \times 8 \times \ldots\ldots \times 98 \times 100$

$= 2^{50} \times (1 \times 2 \times 3 \times \ldots\ldots \times 49 \times 50) = 2^{50} \times 50!$

Clearly, the highest power of 2 in N is much higher than that of 5.

$\therefore$ Number of zeros in N = Highest power of 5 in N =

$\left[\frac{50}{5}\right] + \left[\frac{50}{5^2}\right] = 10 + 2 = 12$.

187. Clearly, the list of prime numbers from 2 to 99 has only 1 multiple of 2 and only 1 multiple of 5.

So, number of zeros in the product = 1.